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MTH 421 — Past Questions & Materials
by Renance

Course: MTH 421
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

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Paper 1 of 1 — Mth 421 Partial Differential Equations 2020 2021 Second Semester

Level 400 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, NIGERIA
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS

B. Sc (MATHEMATICS) DEGREE EXAMINATION (2020/2021), SECOND SEMESTER.
 MTH 421 (PARTIAL DIFFERENTIAL EQUATIONS 3C). September, 2022. Duration: 2hr, 30 mins.

Instructions: Answer ANY FOUR (4) questions.

Credit shall be given to students with brevity and concise answers.

Question 1

(a) Define a first order partial differential equation (PDE) for the function $u(x_1, x_2, x_3, \dots, x_n)$ **3 1/2 marks**

(b) Given a general second order partial differential equation for the unknown $u(x, y)$, with constant

coefficients, $\mathcal{L}u = A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu + G = 0$, **5 marks**

State the classification into 3-major pde category

(c) Given that $xu_x - yu_y + y^2u = y^2, x, y \neq 0$

(i) Determine all characteristic variables to the equations **9 marks**

(ii) Hence find the general solution to the problem

Question 2 A second order partial differential equation (pde) with constant coefficient is given as

$$u_{xx} + 5u_{xy} + 6u_{yy} = 0. \quad \text{4 marks}$$

(a) Classify the pde into any of Parabolic, Elliptic or Hyperbolic pde. **6 1/2 marks**

(b) Reduce the pde to canonical form **6 marks**

(c) Find the general solution of the problem.

Question 3. Consider the wave equation in 1D; $u_{tt} = c^2 u_{xx}$ with $u(x, 0) = f(x), u_t(x, 0) = g(x)$, using characteristic method, **4 marks**

(a) find all the characteristic variable for the 1D pde,

(b) hence or otherwise, show that the D'Alembert solution is **8 marks**

$$u(x, t) = \frac{1}{2} [f(x + ct) + f(x - ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$$

5 1/2 marks

(c) find the solution if $f(x) = \sin 5x$ and $g(x) = \frac{1}{5} \cos x$, **4 marks**

Question 4

(a) Define the Laplace transform of a single valued function $f(x)$

(b) Obtain Laplace transform of the following quantities:

i) $\sin ax$, where a is a constant **6 marks**

ii) $\frac{\partial^2 u}{\partial t^2}(x, t)$

(c) Solve, using Laplace Transform method **7 1/2 marks**

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial t} = x, \quad x > 0, t > 0, \quad u(0, t) = 0 \quad t > 0, \text{ and } u(x, 0) = 0, x > 0.$$

5 Given the first order pde $xu_x - yu_y + y^2u = y^2, x, y \neq 0$. **3 1/2 marks**

(a) Determine all characteristic variables to the pde

(b) Hence or otherwise, find the general solution of the equation. **6 1/2 marks**

(c) Using separation of variables, solve heat conduction and diffusion equation **7 1/2 marks**

$$\frac{\partial u}{\partial t} = 9 \frac{\partial^2 u}{\partial x^2}, u(x, 0) = \sin \pi x, u(0, t) = 0, u(1, t) = 0$$

Question 6 Starting from Taylor series expansion, derive the finite difference approximation to **6 marks**

(a) (i) first - order, and

(ii) second - order derivatives **3 marks**

(b) initial-boundary conditions $u(x, 0) = \sin(\pi x), u_t(x, 0) = 0, 0 \leq x \leq 1, u(0, t) = u(1, t) = 0, t > 0$

(c) Solve the wave equation $u_{tt} = u_{xx}, 0 \leq x \leq 1$, subject to the conditions defined in (b) above.

$u(x, 0) = \sin(\pi x), u_t(x, 0) = 0, 0 \leq x \leq 1, u(0, t) = u(1, t) = 0, t > 0$

using the explicit method with $h = 1/4$ and $k = 1/8$. Compute for four time steps. If the exact solution is $u(x, t) = \cos(\pi t) \sin(\pi x)$, compare the solutions at times $t = 1/4$ and $t = 1/2$. **8 1/2 marks**

Examiner: Prof. A. M. Okedoye

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37
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